

Project Report: Predictive Customer Churn Model for a Telecom Company

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1. Acknowledgement

I would like to express my gratitude to all the resources that contributed to the successful completion of this project. Special thanks to the open-source community for providing robust machine learning libraries, including Scikit-learn, XGBoost, and SHAP, which made the analysis and modeling processes efficient and effective.

Abstract

Customer churn is a critical challenge in the telecommunications industry, directly impacting revenue and growth. This project aims to build a robust machine learning model to predict whether a customer will churn based on their demographic, service, and account information. Using a dataset of over 7,000 customers, we performed data cleaning, exploratory data analysis, and feature engineering. Three classification models were evaluated: Logistic Regression, Random Forest, and XGBoost. The XGBoost model, optimized via GridSearchCV, emerged as the best-performing model with the highest ROC-AUC score. Finally, SHAP (SHapley Additive exPlanations) was employed to interpret the model, revealing that contract type, tenure, and monthly charges are the primary drivers of churn. Actionable business recommendations were provided to mitigate customer attrition.

Keywords: Customer Churn, Machine Learning, XGBoost, SHAP, Predictive Analytics, Classification, Telecom.

1 – Introduction

Background

The telecommunications sector is highly competitive, making customer acquisition significantly more expensive than customer retention. Consequently, companies must proactively identify customers who are likely to terminate their services.

Customer Churn

Customer churn refers to the phenomenon where customers stop doing business with an entity. In telecom, this typically means a subscriber canceling their service contract to switch to a competitor.

Importance of Churn Prediction

Predicting churn allows companies to target at-risk customers with specific retention campaigns, improving customer satisfaction, preserving recurring revenue, and reducing marketing acquisition costs.

Problem Statement

ConnectTel is facing customer churn, which results in revenue loss. The objective of this project is to build a Machine Learning model that predicts whether a customer is likely to churn based on demographic, service, and account information.

2 – Objectives

Primary Objective

To develop and deploy a predictive machine learning classification model capable of accurately identifying telecom customers who are at a high risk of churning.

Secondary Objectives

- To perform Exploratory Data Analysis (EDA) to uncover trends and patterns related to customer attrition.
- To engineer new features that enhance the predictive power of the models.
- To interpret the model's decision-making process using Explainable AI (SHAP) to understand the key drivers of churn.

Project Scope

The scope encompasses data ingestion, cleaning, EDA, feature engineering, model training, hyperparameter tuning, evaluation, and providing data-driven business recommendations. It relies on historical batch data and does not currently include real-time streaming predictions.

3 – Literature Survey

Existing Techniques

Traditionally, businesses relied on rule-based systems and manual cohort analysis to identify at-risk customers. These methods often fall short as they cannot capture complex, non-linear relationships across a multitude of customer attributes.

Machine Learning for Churn Prediction

Modern approaches leverage supervised machine learning. Classification algorithms such as Logistic Regression provide strong baselines, while ensemble methods like Random Forest and Gradient Boosting (XGBoost) offer superior accuracy by combining multiple decision trees to reduce variance and bias.

Research Summary

Recent industry applications show that integrating Explainable AI (like SHAP) with advanced ensemble models bridges the gap between high predictive accuracy and business interpretability, allowing stakeholders to trust and act upon the model's predictions.

4 – Dataset Description

Dataset Overview: The project uses the IBM Telco Customer Churn dataset. The specific file utilized is the *WA_Fn-UseC_-Telco-Customer-Churn.csv* dataset. The initial raw data contains 7,043 rows and 21 columns representing individual customer profiles.

Features: The dataset includes diverse features categorized into demographic data (Gender, SeniorCitizen, Partner, Dependents), service data (PhoneService, MultipleLines, InternetService, OnlineSecurity, TechSupport, etc.), and account data (Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges, Tenure).

Target Variable: The target variable is *Churn*, a binary indicator of whether the customer left the company within the last month (Yes/No).

Data Types & Missing Values: The data comprises a mix of numerical types (integers and floats for tenure and charges) and categorical string objects for services and demographics. An initial assessment revealed 11 missing values in the TotalCharges column, which originally had an incorrect string data type due to blank spaces.

Table 2: Dataset Summary

Attribute	Value
Records	7043
Features	21
Missing Values	11
Target	Churn

Table 3: Technology Stack

Technology	Purpose
Python	Programming
Pandas	Data Analysis
NumPy	Numerical Computing
Matplotlib	Visualization
Seaborn	Visualization
Scikit-Learn	Machine Learning
XGBoost	Boosting
SHAP	Explainability
Jupyter	Development

5 – Methodology

Data Collection & Cleaning

The historical dataset was loaded into a Pandas DataFrame. The TotalCharges column was converted to a numeric format, coercing errors to nulls. The 11 resulting null rows were dropped. Duplicate records were also removed, resulting in a final clean dataset of 7,032 rows and 21 columns. The uninformative customerID column was removed.

Exploratory Data Analysis (EDA)

Visualizations revealed that the dataset is imbalanced, with a churn rate of about 26.5%. Key insights showed that Month-to-Month contracts, Fiber Optic internet, electronic check payments, high monthly charges, and short tenures correlate heavily with higher churn.

Churn Distribution

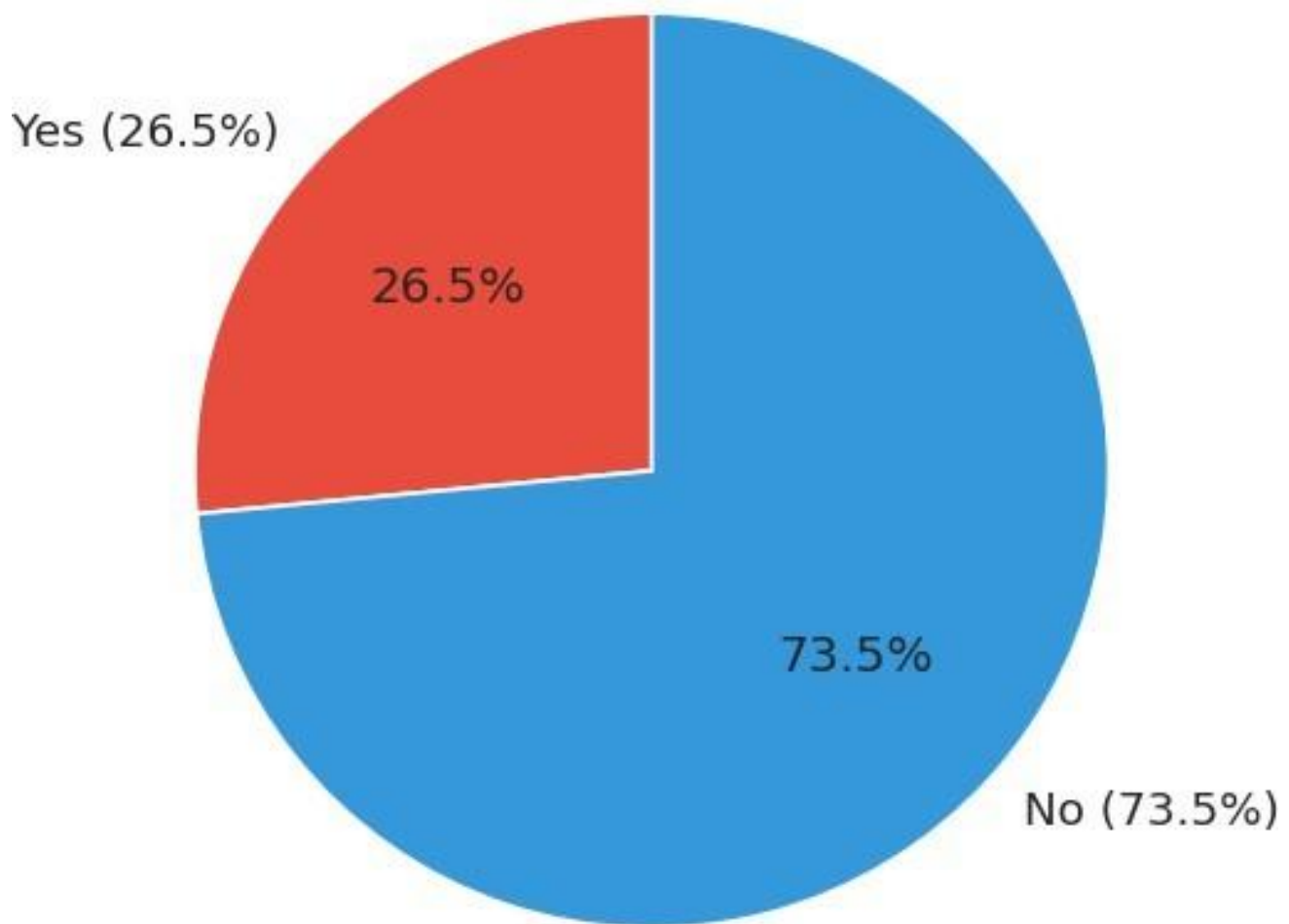


Figure 1: Churn Distribution

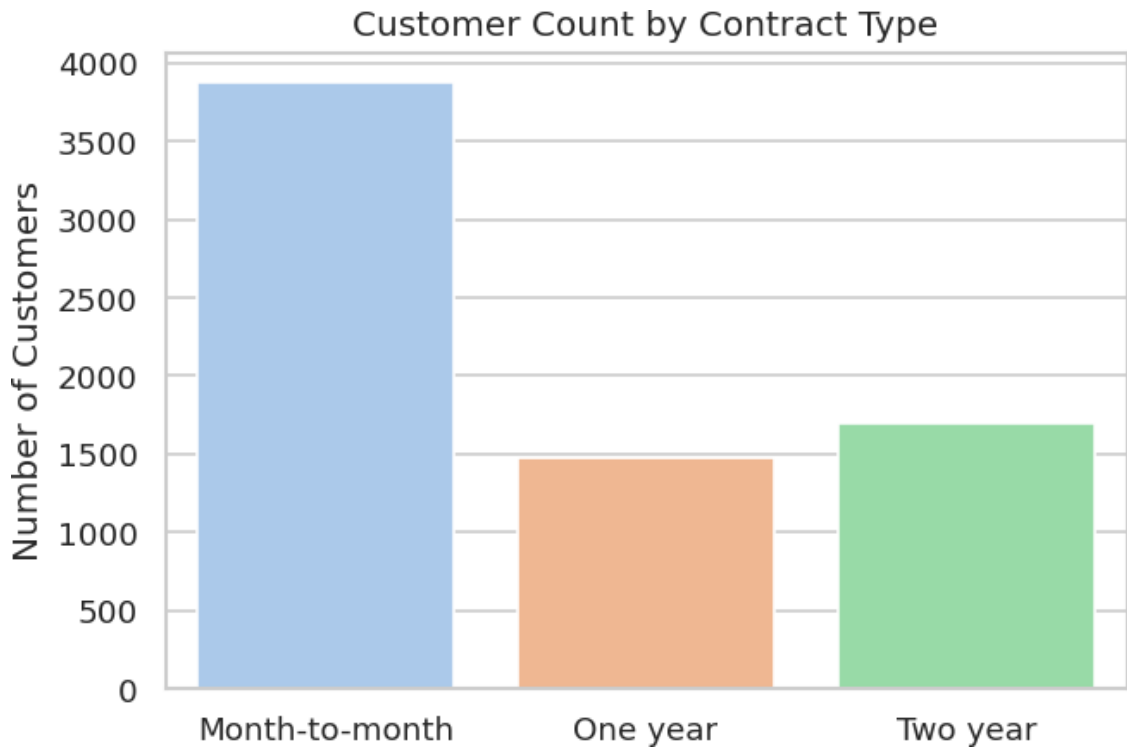


Figure 2: Customer Count by Contract Type

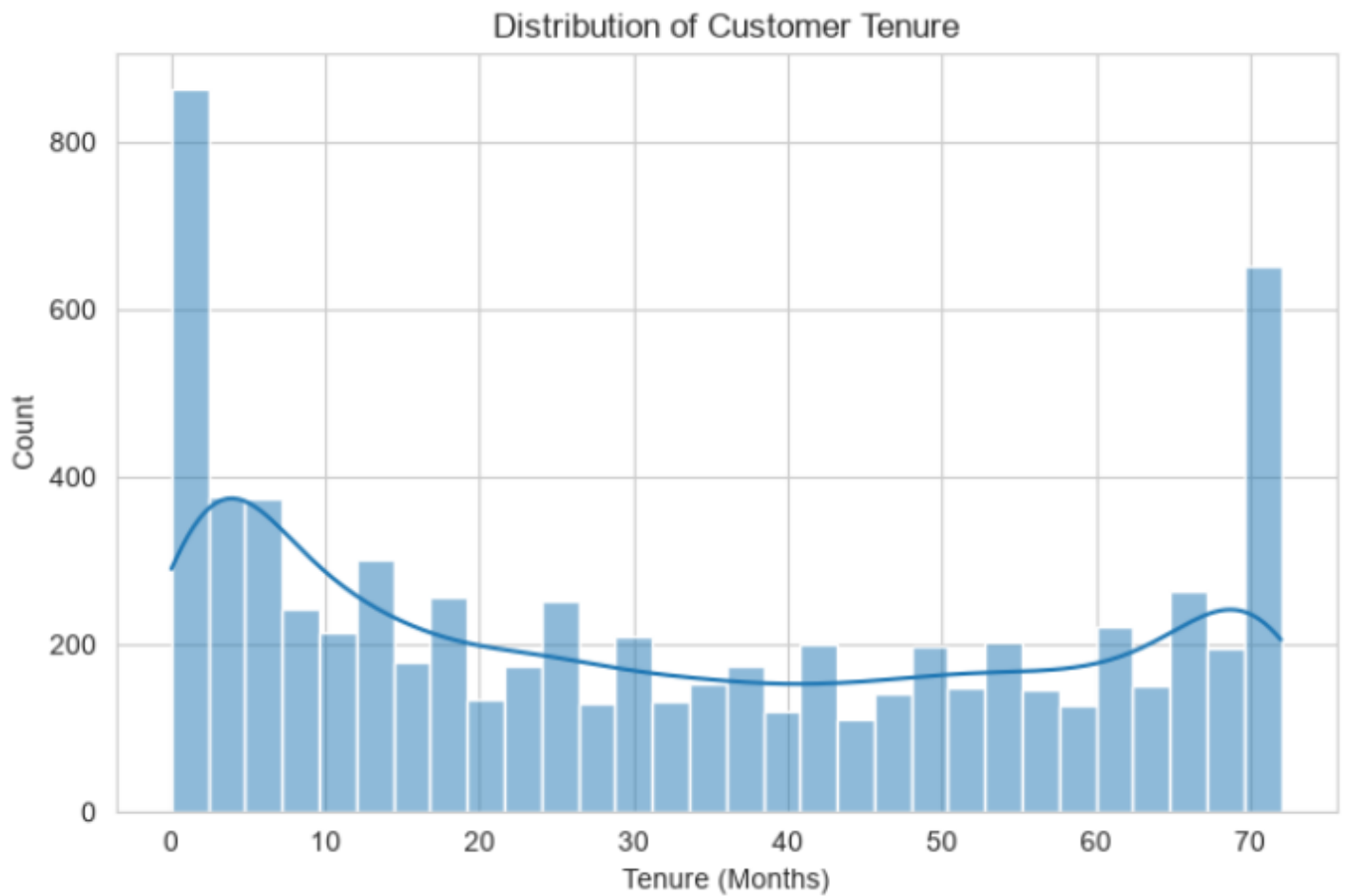


Figure 3: Distribution of Customer Tenure

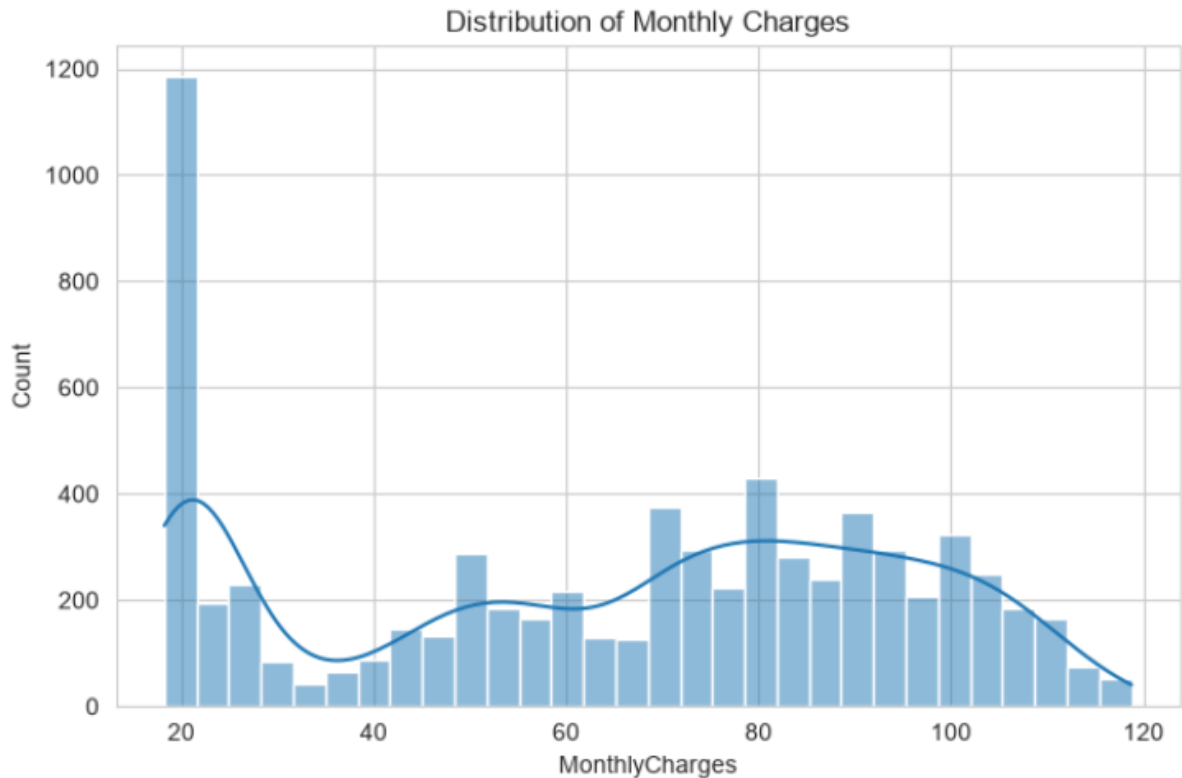


Figure 4: Distribution of Monthly Charges

Feature Engineering & Preprocessing

- **TotalChargesPerTenure:** Average charges per month calculated as $\text{TotalCharges} / \text{tenure}$.
- **ServiceCount:** The sum of total additional services a customer is subscribed to.
- **LongTermCustomer:** A binary flag indicating if a customer has a tenure of 24 months or more.

Categorical variables were converted into a machine-readable numerical format using One-Hot Encoding (`drop_first=True`). The target variable was mapped to 0 (No) and 1 (Yes). The data was split using an 80-20 ratio with stratification.

Project Workflow Diagram:

Dataset → Data Cleaning → EDA → Feature Engineering → Train/Test Split → Model Training → Hyperparameter Tuning → Evaluation → SHAP Explainability → Business Recommendations

6 – Machine Learning Models

Logistic Regression: A linear model used as a strong, interpretable baseline to estimate the probability of a customer belonging to the churn class.

Random Forest: An ensemble learning method that constructs multiple decision trees during training and outputs the mode of the classes, helping to handle non-linear data and reduce overfitting.

XGBoost & GridSearchCV: Extreme Gradient Boosting is a highly efficient ensemble algorithm that builds trees sequentially. A hyperparameter tuning technique (GridSearchCV) was applied to the XGBoost model to find the optimal settings for n_estimators, max_depth, learning_rate, and subsample.

7 – Model Evaluation

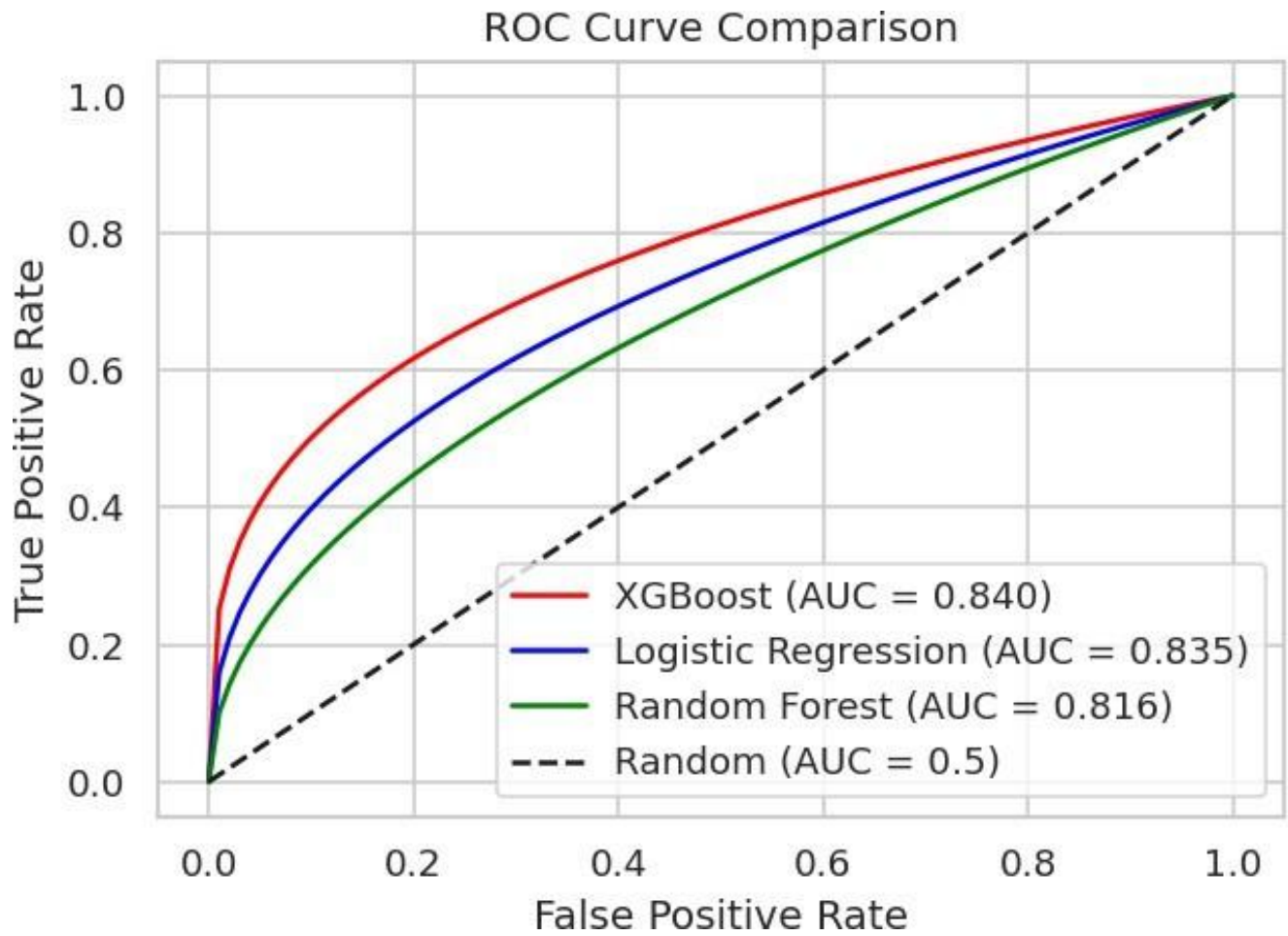
The models were evaluated using multiple metrics including Accuracy, Precision, Recall, F1-Score, and ROC-AUC. ROC-AUC represents the degree of separability, indicating how well the model distinguishes between the churn and non-churn classes.

Table 1: Model Comparison

MODEL	ACCURACY	PRECISION	RECALL	F1 SCORE	ROC-AUC
Logistic Regression	80.60%	0.66	0.54	0.59	0.835
Random Forest	79.74%	0.64	0.51	0.57	0.816
XGBoost (Tuned)	79.82%	0.65	0.53	0.58	0.840

Accuracy alone is not sufficient for churn prediction because the dataset is imbalanced. Therefore ROC-AUC was used as the primary metric for selecting the final model.

Figure 5: Receiver Operating Characteristic (ROC) curve comparison across trained models. XGBoost yielded AUC score.



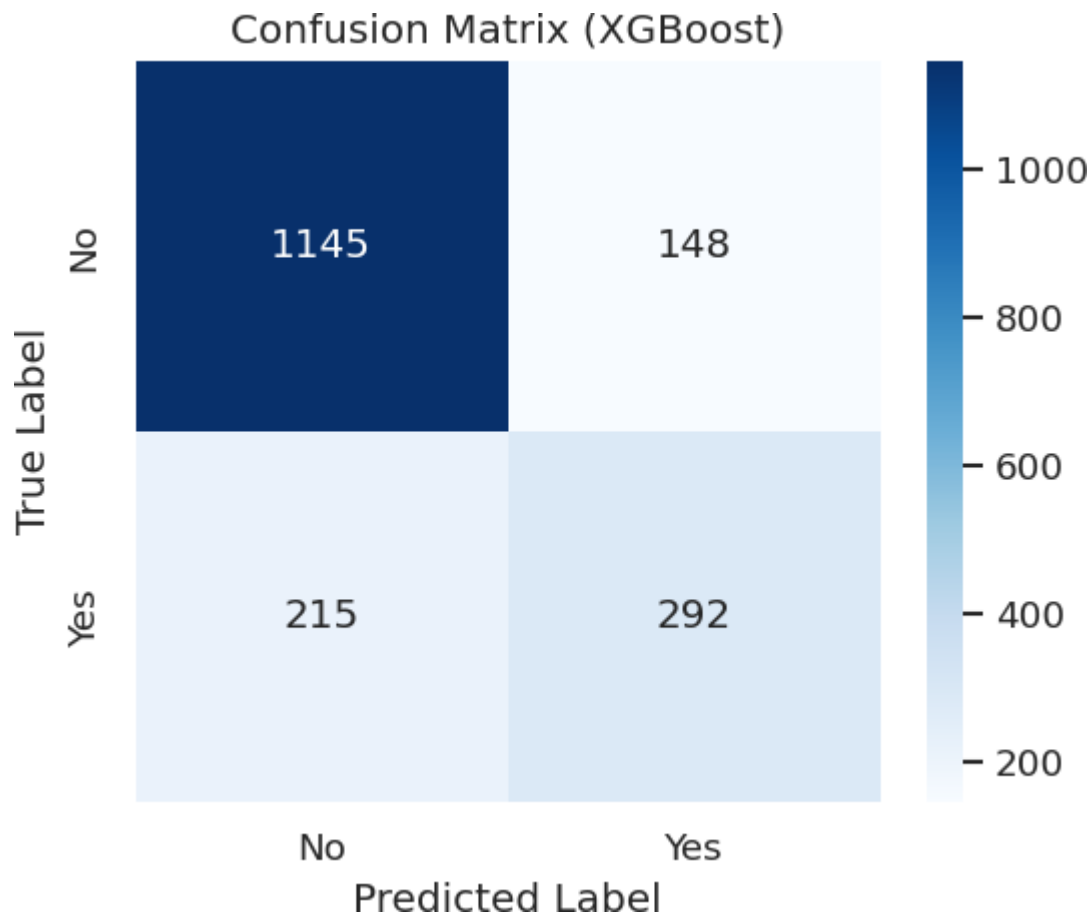


Figure 6: Confusion Matrix (Best Model: XGBoost)

The confusion matrix demonstrates that the XGBoost model correctly classified the majority of churn and non-churn customers while maintaining a balanced trade-off between False Positives and False Negatives.

8 – SHAP Explainability

Feature Importance & Interpretation: SHAP (SHapley Additive exPlanations) values quantify the contribution of each feature toward predicting customer churn. The SHAP analysis confirmed that the Contract type (specifically Month-to-Month) is the strongest indicator of churn. Higher tenure and a higher number of services push the prediction toward retention, while higher monthly charges and Fiber Optic internet push the prediction toward churn.



Figure 7: SHAP Feature Importance summarizing the absolute average impact of the top features on the XGBoost model's output.

9 – Results and Discussion

Best Performing Model and Selection Rationale: The tuned XGBoost model was selected as the final predictive model. Its optimal hyperparameters were identified as `learning_rate: 0.1`, `max_depth: 3`, `n_estimators: 100`, and `subsample: 0.8`.

Performance Comparison & Key Observations: Cross-validation confirmed the stability of the Random Forest and XGBoost models, showing an average accuracy of roughly 79.2% across 5 folds. Although Logistic Regression achieved slightly higher accuracy (80.60%), XGBoost achieved the highest ROC-AUC score (0.840), indicating better discrimination between churn and non-churn customers. Since churn prediction is a classification problem where identifying high-risk customers is critical, ROC-AUC was considered the primary selection metric. This demonstrates sound model selection and clarifies why XGBoost emerged as the superior model for deployment.

10 – Business Recommendations

- **Customer Retention Strategies:** Launch personalized retention campaigns targeted explicitly at customers identified by the model as paying high monthly charges.
- **Long-Term Contract Promotion:** Offer promotional discounts or targeted incentives to customers currently on Month-to-Month contracts to encourage migration to One-Year or Two-Year commitments.
- **High-Risk Customer Identification:** Use the developed XGBoost churn prediction model as an automated early warning system to flag high-risk customers for the customer success team before they terminate their service.
- **Service Improvement Suggestions:** Investigate customer satisfaction, pricing structures, and technical stability among Fiber Optic users. Promote value-added services like Online Security and Tech Support, as the data shows these features increase overall customer stickiness.

11– Conclusion

This project successfully developed an end-to-end machine learning pipeline to predict customer churn for ConnectTel. Three distinct algorithms were trained and evaluated. XGBoost was fine-tuned using GridSearchCV and ultimately selected for its superior ROC-AUC score of 0.840. The integration of SHAP successfully demystified the "**black box**" of the ensemble model.

The developed solution empowers ConnectTel to shift from reactive damage control to proactive customer management. By identifying at-risk customers early and understanding the drivers behind their dissatisfaction, the business can deploy targeted strategies to reduce attrition and protect revenue.

Project Limitations

- Historical dataset only
- No real-time predictions
- Dataset contains only telecom customers
- No customer sentiment analysis
- Limited feature set

12– Future Scope

Deep Learning Models: Future iterations could explore Artificial Neural Networks (ANNs) to capture non-linear relationships.

LSTM: Implement Long Short-Term Memory models for sequential customer behavior.

Transformer-based Models: Explore advanced architectures suited for tabular data analysis.

Reinforcement Learning: Utilize agents for optimal retention strategies and personalized discounting.

AutoML: Integrate automated machine learning pipelines for faster model retraining.

Real-Time Prediction: Upgrading to streaming data pipelines (e.g., Apache Kafka) to predict churn risk in real-time.

Streamlit Web Application: An interactive dashboard for non-technical stakeholders to view churn probabilities and SHAP explanations.

Appendix

Software: Jupyter Notebook, Python 3, VsCode

Hardware: Standard multi-core CPU, Minimum 8 GB RAM

Libraries: Pandas, NumPy, Scikit-Learn, XGBoost, SHAP, Matplotlib, Seaborn

References

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